

S.No	Particulars	Theory		Practical		
		Amplitude	Frequency	Amplitude	Time	Frequency
1.	Message Signal	3V	300 Hz	3V	$T_m = X_2 - X_1$ $= 0.00335$	300 Hz
2.	Carrier Signal	5V	3000 Hz	5V	$T_m = X_2 - X_1$ $= 0.000333$	3000 Hz
3.	AM wave	-	-	$E_{max} = 8V$ $E_{min} = 2V$	$T_{AM} = X_2 - X_1$ $= 0.000333$	3000 Hz

Exo 1.
21/07/2026

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SIMULATION OF AMPLITUDE MODULATION (AM) AND DEMODULATION OF MODULATOR INDEX AND BAND WIDTH ANALYSIS

Example spectrum

with limiter

AIM:

To generate and analysis an amplitude modulation (AM) wave using SCI LAB and observe the message signal, carrier signal and wave form.

Apparatus Required

- * Laptop with i5 processor
- * SCI LAB Software.

CODE:

```
clc;  
clear all;  
close;  
Em = 3;  
Fm = 300;  
EC = 5;  
FC = 3000;  
Fs = 300000;  
t = 0: 1/Fs : 3/Fm;  
Lm = Em * sin(2 * %pi * Fm * t);  
subplot(3, 1, 1);  
plot(t, Lm)  
Lc = em * sin(2 * %pi * fc * t);
```

Calculation:

1. Message signal:

Practical time

$$T_m = x_2 - x_1$$

$$x_2 = 0.00333 \text{ s}$$

$$x_1 = 0 \text{ s}$$

$$T_m = 0.00333 \text{ s}$$

Practical Frequency:

$$f_m = 1/T_m$$

$$= 1/0.00333$$

$$= 300 \text{ Hz}$$

2. Carrier Signal

Practical Time

$$T_c = x_2 - x_1$$

$$x_2 = 0.00333 \text{ s}$$

$$x_1 = 0 \text{ s}$$

$$T_c = 0.00333 \text{ s}$$

Practical Frequency

$$f_c = 1/T_c$$

$$f_c = \frac{1}{0.00333}$$

$$f_c = 3000 \text{ Hz}$$

Sub plot (3, 1, 2);
plot (t, Ie);

Long 2 HD 3

$$I_{am} = (I_m + E_c) \cdot \cos(2 \cdot \pi \cdot f_c \cdot t);$$

Sub plot (3, 1, 3);

plot (t, Iam);

$$E_{max} = E_c + E_m$$

$$V_8 = \dots$$

$$E_{min} = E_c - E_m$$

$$V_8 = \dots$$

Practical Time:

$$T_{am} = X_2 - X_1$$

$$X_2 = (0.000332)$$

$$X_1 = 0.000132$$

$$T_{am} = 0.000200000.0$$

Practical Frequency

$$T_{am} = 1/T_{am}$$

$$1/0.000200000.0$$

$$T_{am} = 3.20 \text{ Hz}$$

3. AM Signal

Practical Amplitude

$$E_{\max} = E_c + E_m$$

$$E_{\max} = 8V$$

$$E_{\min} = E_c - E_m$$

$$= 2V$$

Practical Time :-

$$T_{\text{am}} = x_2 - x_1$$

$$x_2 = (0.000333s)$$

$$x_1 = 0.000113s$$

$$T_{\text{am}} = 0.000327868s$$

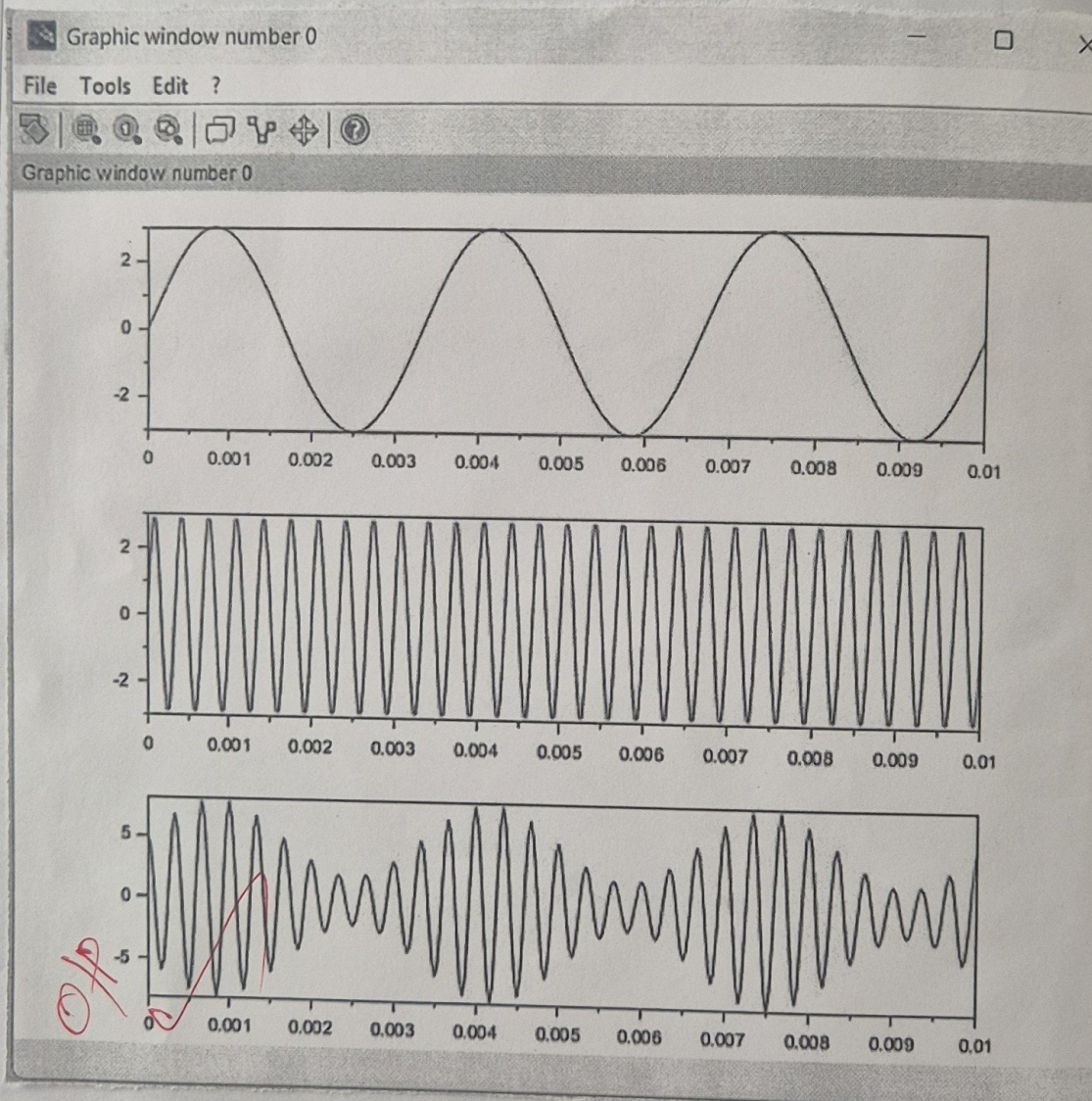
Practical Frequency

$$f_{\text{am}} = 1/T_{\text{am}}$$

$$= 1/0.000327689$$

$$f_{\text{am}} = 3050\text{Hz}$$

Output.



Sno	Content	Mark	Mark obtained
1.	Program implitation	20	20
2.	wave form & Tabulation	20	20
3.	Calculation and result	20	20 15
4.	Via voice	20	20
5.	Result	20	
		Total	95

Result:-

Thus the Amplitude Modulation (AM) signal
 was successfully generated using SET LAB
 28/8/20